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UTM Faculty of Computing
UNIVERSITI TEKNOLOGI MALAYSIA

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER 1, 2024/2025

SUBJECT CODE	**: SECJ3323**
SUBJECT NAME	**: SOFTWARE DESIGN & ARCHITECTURE**
DATE / DAY	**:**
DURATION	**: 3 HOURS**
VENUE	**:**

INSTRUCTIONS

Answer all questions in the Answer Booklet. All diagrams must be drawn using the UML standard notations by hand. Note that any form of exam misconduct caught during the exam will cause students to get a zero in this assessment. This question booklet needs to be returned with the Answer Booklet and Answer Sheets (if any) by the end of the examination. You are only allowed to open this booklet with the permission of the examination invigilators' permission.

(Please Write Your Lecturer Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **6** printed pages excluding this page.

SECTION A - STRUCTURED QUESTIONS

INSTRUCTIONS: Answer ALL of the following questions.

Question 1 **(16 marks)**

a) Assumed that you have five team members to complete the project based on the given case study in Figure 1. Draw a WBS (Work Breakdown Structure) with a linear responsibility chart and distribute among five of you. The proposed WBS must achieve the completeness criteria to ensure that the WBS include the complete set of tasks necessary to complete the project.

A university wants to develop a student portal that serves as a central hub for students to access their academic records, assignments, course materials, and university announcements. The portal must support various user roles with personalized dashboards and secure access to resources based on each role.

Figure 1: Case Study University Student Portal

(6 Marks)

b) You are managing a software design project composed of TWENTY (20) tasks. Each task is estimated to cost $1,000, and the total Budget at Completion (BAC) is $20,000. The project is planned to take TWENTY (20) months, with each task representing 5% of the total project. In the current 6th month, only FIVE (5) tasks have been fully completed at a cost of $8,000. Using Earned Value Management (EVM) provided in the appendix, calculate all necessary value needed.

(6 Marks)

c) There are FIVE (5) personality dimensions that categorize groups of traits. The idea is to be able to classify someone's personality into one of the dimensions. Based on your own experience, explain how you apply TWO (2) personality dimensions during your project in this course.

(4 Marks)

Question 2 **(12 marks)**

a) What is Programming Design Language (PDL), and how is it used in software development?
(2 Marks)

b) List TWO (2) key quality attributes that are commonly evaluated in software design construction to ensure the quality of the design.
(2 Marks)

The Customer Service Process in a Hotel Management System begins with customer inquiries about room availability, rates, and amenities. If rooms are available, the customer proceeds with the reservation process, during which they can make an advance payment or choose to pay upon arrival. Upon check-in, the system verifies the booking details and collects payment if it was not made earlier. During their stay, customers can request additional services, such as room cleaning, food delivery, or maintenance, which the system tracks and facilitates. At check-out, the system calculates any remaining charges and ensures all payments are settled. Before closing the booking, the system may optionally collect customer feedback or reviews to improve service quality.

Figure 2: Customer Service Process

c) Based on the Figure 2 above Hotel Management System Description document on Customer Service process, draw an EEDT (Extended Entry Decision Table).
(8 Marks)

Question 3 **(12 marks)**

Answer the following questions based on sample code in Figure 3.

```
1   import java.util.Scanner;
2   public class CharityEligibility {
3    public static void main(String[] args) {
4    Scanner scanner = new Scanner(System.in);
5    // Input number of family members
6    int numberOfMembers;
7    do {
8     System.out.print("Enter the number of family members (up to 10): ");
9             numberOfMembers = scanner.nextInt();
10    } while (numberOfMembers < 1 || numberOfMembers > 10);
11
12    // Input salaries
13    double[] salaries = new double[numberOfMembers];
14    double totalSalaries = 0;
15    for (int i = 0; i < numberOfMembers; i++) {
16    System.out.print("Enter salary for family member" +(i + 1) + ": RM ");
17              salaries[i] = scanner.nextDouble();
18              totalSalaries += salaries[i];
19     }
```

```
20      // Input expenses
21      double[] expenses = new double[numberOfMembers];
22      double householdExpenses = 0;
23      double totalExpenses = 0;
24      for (int i = 0; i < numberOfMembers; i++) {
25      System.out.print("Enter expenses for family member" +(i + 1)+":RM ");
26              expenses[i] = scanner.nextDouble();
27              totalExpenses += expenses[i];
28       }
29       // Input household expenses
30       System.out.print("Enter household expenses: RM ");
31       householdExpenses = scanner.nextDouble();
32       totalExpenses += householdExpenses;
33    // Check eligibility
34    if (totalSalaries >= totalExpenses || (totalExpenses / numberOfMembers) >
35       1500 || householdExpenses > (0.5 * totalExpenses)) {
36       System.out.println("Sorry, you are not eligible to accept charity.");
37       } else {
38           boolean eligible = true;
39           for (int i = 0; i < numberOfMembers; i++) {
40           if (expenses[i] > 2000) {
41             eligible = false;
42               break;    }  }
43           if (!eligible) {
44       System.out.println("Sorry, you are not eligible to accept charity.");
45                } else {
46       double charityAmount = Math.min(totalExpenses - totalSalaries, 2000);
47    System.out.println("You are eligible to accept charity. Charity amount: RM
48       " + charityAmount);  }
49    }
50    scanner.close();}}
51
```

Figure 3: Sample Code of Charity Eligibility

a) Draw a flowchart to represent the flow of the algorithm based on Figure 3.

(8 Marks)

b) Based on Figure 3, determine the cyclomatic complexity using any approach.

(4 Marks)

SECTION B - PROBLEM SOLVING

INSTRUCTIONS: Answer ALL the following questions.

Question 4 **(30 marks)**

Project 1: Online Payment Processing System

An online payment processing system needs to handle different payment methods, such as Credit Card, PayPal, and Bank Transfer. The system should allow for adding new payment methods without modifying the core logic. Each payment type processes transactions differently, but all share the following steps: validating the payment method, initiating the transaction, and completing it.

Project 2: Cross-Platform UI Component Factory

A software company develops cross-platform applications for both Windows and macOS. The application requires UI components like Buttons, Checkboxes, and TextFields that look native on each platform. Each platform's components have their own attributes and rendering methods.

a) Analyze the scenario of both project and identify a suitable Creational detail design pattern for each project above. Justify for both chosen Creational detail design pattern.

(6 marks)

b) Choose ONE (1) project and design the class diagram for the Creational detail design pattern. The class diagram must reflect your Creational detail design pattern answered in (a).

(10 marks)

c) Elaborate your diagram in (b), describe the components of the diagram**.**

(4 marks)

d) Also based on the class diagram answered in (b), evaluate and write the code representing the class diagram.

(10 marks)

Question 5 **(30 marks)**

Project 1: Online Payment Processing System

A stock market application allows users to subscribe to updates for specific stocks. When the stock price changes, all subscribed users are notified. The system needs to support adding or removing subscriptions dynamically.

Project 2: Media Player Application

A media player application supports playing audio and video files. Initially, it supports .mp3 and .mp4 formats. A new requirement adds .avi file support, but the existing system doesn't directly support .avi. The system should use an adapter to handle the new format seamlessly.

a) Analyze the scenario of both project and identify a suitable Behavioral detail design pattern for each project above. Justify for both chosen Behavioral detail design pattern.

(6 marks)

b) Choose ONE(1) project and design the class diagram for the Behavioral detail design pattern. The class diagram must reflect your Behavioral detail design pattern answered in (a).

(10 marks)

c) Elaborate your diagram in (b), describe the components of the diagram**.**

(4 marks)

d) Also based on the class diagram answered in (b), evaluate, and write the code representing the class diagram.

(10 marks)

- END OF QUESTIONS -

APPENDIX

Parameter	Description
EV	Also called *budgeted cost of work performed (BCWP)*, it is the dollar amount of the work currently completed
AC	Also called *actual cost of work performed (ACWP)*, it represents the amount of money that was spent in the work currently completed
PV	Also called *budgeted cost of work scheduled (BCWS)*, it represents the value of the work that was expected to be completed at that point in time of the schedule
CV	Describes the difference between the value of work completed and the actual cost spent to complete it
SV	Describes the difference between the value of work currently completed and the cost of the work that was expected to be completed
CPI	Describes the current cost efficiency of the work (CPI < 1 means over budget; CPI > 1 means under budget; CPI = 1 means right on budget)
SPI	Describes the current schedule efficiency of the work (SPI < 1 means behind schedule; SPI > 1 means ahead of schedule; SPI = 1 means right on schedule)
BAC	Total budget for the project
ETC	Represents the estimated cost remaining to complete the project
EAC	Represents is the total cost of the entire project

Parameter	Equation
EV	EV = (% work completed) * planned cost for the work
CV	CV = EV - AC
SV	SV = EV - PV
CPI	CPI = EV / AC
SPI	SPI = EV / PV
ETC	ETC = (BAC – EV) / CPI
EAC	EAC = ETC + AC